

Chronic intake of potato chips in humans increases the production of reactive oxygen radicals by leukocytes and increases plasma C-reactive protein...

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BACKGROUND: Relatively high concentrations of acrylamide in commonly ingested food products, such as French fries, potato chips, or cereals, may constitute a potential risk to human health. OBJECTIVE: The objective of this pilot study was to investigate the possible connection between chronic ingestion of acrylamide-containing potato chips and oxidative stress or inflammation. DESIGN: Fourteen healthy volunteers (mean age: 35 y; 8 women and 6 smokers of >20 cigarettes/d) were given 160 g of potato chips containing 157 microg [corrected] acrylamide daily for 4 wk. RESULTS: An increase in acrylamide-hemoglobin adducts in blood was found in all the study subjects, with a mean of 43.1 pmol x L(-1) x g(-1) hemoglobin (range: 27-76; P < 0.01) in nonsmokers and 59.0 pmol x L(-1) x g(-1) hemoglobin (range: 43-132; P < 0.05) in smokers. Concurrently, a significant increase (P < 0.01) in the oxidized LDL, high-sensitivity interleukin-6, high-sensitivity C-reactive protein, and gamma-glutamyltransferase concentrations was observed in both smokers and nonsmokers. A significant increase in reactive oxygen radical production by monocytes, lymphocytes, and granulocytes and an increase in CD14 expression in macrophages (P < 0.001) were found after intake of potato chips. Twenty-eight days from the discontinuation of the experiment, the variables under study decreased to some extent. It has been shown also that acrylamide increases the production of reactive oxygen species in isolated human monocyte-macrophages in vitro and decreases the cellular glutathione concentration. CONCLUSION: These novel findings seem to indicate that chronic ingestion of acrylamide-containing products induces a proinflammatory state, a risk factor for progression of atherosclerosis.